

Let loss = L, inverse sum = I, sum = C_s and counts = C. P = C * I.

1. Prove “dcounts_sum_inv = (counts * dprobs).sum(1, keepdim=True)”

$$\frac{\partial L}{\partial I} = \sum_i \frac{\partial L}{\partial P_i} \times \frac{\partial P_i}{\partial I} = \sum_i \frac{\partial L}{\partial P_i} C_i$$

Although the loss only considers a particular probability, that probability actually is influenced by the neighbouring distribution. Therefore, Loss is dependent on all P_i.

2. Prove “dcounts = counts_sum_inv * dprobs + torch.ones_like(counts) * dcounts_sum”

$$\begin{aligned} \frac{\partial L}{\partial C_i} &= \frac{\partial L}{\partial P_i} \times \frac{\partial P_i}{\partial C_i} \\ &= \frac{\partial L}{\partial P_i} \times \left(I + C_i \frac{\partial I}{\partial C_i} \right) \\ &= \frac{\partial L}{\partial P_i} I + \frac{\partial L}{\partial P_i} \frac{\partial P_i}{\partial I} \frac{\partial I}{\partial C_s} \times \frac{\partial C_s}{\partial C_i} = \frac{\partial L}{\partial P_i} I + \frac{\partial L}{\partial C_s} \end{aligned}$$

3. Prove the loss gradient of linear transformation (Y = AX + B)

In tensor form: $y_{ij} = A_{ik}X_{kj} + B_j$, B_j value is independent on i .

Case 1:

$$\begin{aligned} \frac{\partial L}{\partial A_{ik}} &= \sum_{\alpha, \beta} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial y_{\alpha\beta}}{\partial A_{ik}} = \sum_{\alpha, \beta, \gamma} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial (A_{\alpha\gamma}X_{\gamma\beta} + B_\beta)}{\partial A_{ik}} \\ &= \sum_{\alpha, \beta, \gamma} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial A_{\alpha\gamma}}{\partial A_{ik}} X_{\gamma\beta} = \sum_{\alpha, \beta, \gamma} \frac{\partial L}{\partial y_{\alpha\beta}} \delta_{i\alpha} \delta_{k\gamma} X_{\gamma\beta} \\ &= \sum_{\beta} \frac{\partial L}{\partial y_{i\beta}} X_{k\beta} = \sum_j \frac{\partial L}{\partial y_{ij}} [X^T]_{jk} \end{aligned}$$

Case 2:

$$\begin{aligned} \frac{\partial L}{\partial X_{kj}} &= \sum_{\alpha, \beta} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial y_{\alpha\beta}}{\partial X_{kj}} = \sum_{\alpha, \beta, \gamma} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial (A_{\alpha\gamma}X_{\gamma\beta} + B_\beta)}{\partial X_{kj}} = \sum_{\alpha, \beta, \gamma} \frac{\partial L}{\partial y_{\alpha\beta}} A_{\alpha\gamma} \delta_{j\beta} \delta_{k\gamma} \\ &= \sum_{\alpha} \frac{\partial L}{\partial y_{\alpha j}} A_{\alpha k} = \sum_i [A^T]_{ki} \frac{\partial L}{\partial y_{ij}} \end{aligned}$$

Case 3:

$$\frac{\partial L}{\partial B_j} = \sum_{\alpha, \beta} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial y_{\alpha\beta}}{\partial B_j} = \sum_{\alpha, \beta} \frac{\partial L}{\partial y_{\alpha\beta}} \frac{\partial B_\beta}{\partial B_j} = \sum_i \frac{\partial L}{\partial y_{ij}}$$

4. Prove the gradient of “hpreact = bngain * bnraw + bnbias”.

Let $z_{ij} = G_j M_{ij} + B_j$; noted that summation convention is not applied this time.

$$\begin{aligned}\frac{\partial L}{\partial G_j} &= \sum_{i,k} \frac{\partial L}{\partial z_{ik}} \frac{\partial z_{ik}}{\partial G_j} = \sum_{i,k} \frac{\partial L}{\partial z_{ik}} \frac{\partial (G_k M_{ik} + B_k)}{\partial G_j} \\ &= \sum_{i,k} \frac{\partial L}{\partial z_{ik}} M_{ik} \delta_{jk} = \sum_i \frac{\partial L}{\partial z_{ij}} M_{ij}\end{aligned}$$

$$\frac{\partial L}{\partial B_j} = \sum_i \frac{\partial L}{\partial z_{ij}}$$

Same as previous

$$\begin{aligned}\frac{\partial L}{\partial M_{ij}} &= \sum_{\alpha,\beta} \frac{\partial L}{\partial z_{\alpha\beta}} \frac{\partial z_{\alpha\beta}}{\partial M_{ij}} = \sum_{\alpha,\beta} \frac{\partial L}{\partial z_{\alpha\beta}} \frac{\partial (G_\beta M_{\alpha\beta} + B_\beta)}{\partial M_{ij}} \\ &= \sum_{\alpha,\beta} \frac{\partial L}{\partial z_{\alpha\beta}} G_\beta \delta_{\alpha i} \delta_{\beta j} = \frac{\partial L}{\partial z_{ij}} G_j\end{aligned}$$

5. Prove the gradient of “bnraw = bndiff * bnvar_inv” of bndiff

Let $Z_{ij} = X_{ij} Y_j$, where $Y_j = Y(X_{ij})$.

$$\begin{aligned}\frac{\partial L}{\partial X_{ij}} &= \sum_{\alpha,\beta} \frac{\partial L}{\partial Z_{\alpha\beta}} \frac{\partial Z_{\alpha\beta}}{\partial X_{ij}} = \sum_{\alpha,\beta} \frac{\partial L}{\partial Z_{\alpha\beta}} \frac{\partial (X_{\alpha\beta} Y_\beta)}{\partial X_{ij}} \\ &= \sum_{\alpha,\beta} \frac{\partial L}{\partial Z_{\alpha\beta}} \left(\frac{\partial X_{\alpha\beta}}{\partial X_{ij}} Y_\beta + X_{\alpha\beta} \frac{\partial Y_\beta}{\partial X_{ij}} \right) \\ &= \frac{\partial L}{\partial Z_{ij}} Y_j + \sum_{\alpha,\beta} \frac{\partial L}{\partial Z_{\alpha\beta}} \frac{\partial Z_{\alpha\beta}}{\partial Y_\beta} \frac{\partial Y_\beta}{\partial X_{ij}}\end{aligned}$$

6. Prove dloss / dlogits.

$$\begin{aligned}L &= -\frac{1}{N} \sum_{n=0}^{N-1} \ln \left(\frac{e^{l_{n,n_i}}}{\sum_j e^{l_{n,n_j}}} \right) \\ \frac{\partial L}{\partial l_{m,m_k}} &= -\frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_j e^{l_{n,n_j}}}{e^{l_{n,n_i}}} \frac{\partial}{\partial l_{m,m_k}} \frac{e^{l_{n,n_i}}}{\sum_j e^{l_{n,n_j}}} \\ &= -\frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_j e^{l_{n,n_j}}}{e^{l_{n,n_i}}} \frac{1}{\left(\sum_j e^{l_{n,n_j}} \right)^2} \left(\frac{\partial e^{l_{n,n_i}}}{\partial l_{m,m_k}} \sum_j e^{l_{n,n_j}} - e^{l_{n,n_i}} \frac{\partial \sum_j e^{l_{n,n_j}}}{\partial l_{m,m_k}} \right) \\ &= -\frac{1}{N} \sum_{n=0}^{N-1} \frac{1}{e^{l_{n,n_i}} \sum_j e^{l_{n,n_j}}} \left(e^{l_{n,n_i}} \delta_{mn} \delta_{ik} \sum_j e^{l_{n,n_j}} \right. \\ &\quad \left. - e^{l_{n,n_i}} \sum_j e^{l_{n,n_j}} \delta_{mn} \delta_{jk} \right)\end{aligned}$$

$$\frac{\partial L}{\partial l_{m,m_k}} = -\frac{1}{N} \frac{1}{\sum_j e^{l_{m,m_j}}} \left(\delta_{ik} \sum_j e^{l_{m,m_j}} - e^{l_{m,m_k}} \right) = \frac{1}{N} \left(\frac{e^{l_{m,m_k}}}{\sum_j e^{l_{m,m_j}}} - \delta_{ik} \right)$$

7. Prove d loss / d hprebn

$$Z_{n,n_i} = \gamma_i \frac{x_{n,n_i} - \mu_i}{\sigma_i} + b_i \equiv \gamma_i \hat{x}_{n,n_i} - \mu_i$$

, where μ_i and σ_i are dependent on x_{n,n_i} .

$$\frac{\partial L}{\partial x_{m,m_j}} = \sum_n \frac{\partial L}{\partial Z_{n,n_i}} \frac{\partial}{\partial x_{m,m_j}} \left(\gamma_i \frac{x_{n,n_i} - \mu_i}{\sigma_i} + b_i \right)$$

$$\frac{\partial}{\partial x_{m,m_j}} \left(\frac{x_{n,n_i} - \mu_i}{\sigma_i} + b_i \right) = \frac{1}{\sigma_i^2} \left[\frac{\partial (x_{n,n_i} - \mu_i)}{\partial x_{m,m_j}} \sigma_i - (x_{n,n_i} - \mu_i) \frac{\partial \sigma_i}{\partial x_{m,m_j}} \right]$$

Because

$$\mu_i = \frac{1}{N} \sum_n x_{n,n_i} \Rightarrow \frac{\partial \mu_i}{\partial x_{m,m_j}} = \frac{1}{N} \sum_n \frac{\partial x_{n,n_i}}{\partial x_{m,m_j}} = \frac{1}{N} \sum_n \delta_{mn} \delta_{ij} = \frac{1}{N} \delta_{ij}$$

$$\sigma_i = \sqrt{\frac{1}{N-1} \sum_n (x_{n,n_i} - \mu_i)^2}$$

$$\begin{aligned} \frac{\partial \sigma_i}{\partial x_{m,m_j}} &= \frac{1}{2\sigma_i} \frac{1}{N-1} \sum_n 2(x_{n,n_i} - \mu_i) \frac{\partial (x_{n,n_i} - \mu_i)}{\partial x_{m,m_j}} \\ &= \frac{1}{N-1} \frac{1}{\sigma_i} \sum_n (x_{n,n_i} - \mu_i) \left(\delta_{mn} \delta_{ij} - \frac{1}{N} \delta_{ij} \right) \\ &= \frac{1}{N-1} \frac{1}{\sigma_i} \sum_n \left(x_{n,n_i} \delta_{mn} \delta_{ij} - \frac{1}{N} x_{n,n_i} \delta_{ij} - \mu_i \delta_{mn} \delta_{ij} + \frac{1}{N} \mu_i \delta_{ij} \right) \\ &= \frac{1}{N-1} \frac{1}{\sigma_i} \left[x_{m,m_i} \delta_{ij} - \frac{1}{N} \sum_n x_{n,n_i} \delta_{ij} - \mu_i \delta_{ij} + \mu_i \delta_{ij} \right] \\ &= \frac{1}{N-1} \frac{x_{m,m_i} - \mu_i}{\sigma_i} \delta_{ij} = \frac{1}{N-1} \hat{x}_{m,m_i} \delta_{ij} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{\partial}{\partial x_{m,m_j}} \left(\frac{x_{n,n_i} - \mu_i}{\sigma_i} \right) &= \frac{1}{\sigma_i^2} \left[\left(\delta_{mn} \delta_{ij} - \frac{1}{N} \delta_{ij} \right) \sigma_i - (x_{n,n_i} - \mu_i) \frac{1}{N-1} \hat{x}_{m,m_i} \delta_{ij} \right] \\ &= \frac{1}{\sigma_i} \left[\left(\delta_{mn} - \frac{1}{N} \right) - \frac{1}{N-1} \hat{x}_{m,m_i} \hat{x}_{n,n_i} \right] \delta_{ij} \\ \frac{\partial L}{\partial x_{m,m_j}} &= \gamma_i \sum_n \frac{\partial L}{\partial Z_{n,n_i}} \frac{1}{\sigma_i} \left[\left(\delta_{mn} - \frac{1}{N} \right) - \frac{1}{N-1} \hat{x}_{m,m_i} \hat{x}_{n,n_i} \right] \delta_{ij} \\ &= \frac{\gamma_i}{N \sigma_i} \delta_{ij} \left[N \frac{\partial L}{\partial Z_{m,m_i}} - \sum_n \frac{\partial L}{\partial Z_{n,n_i}} - \frac{N}{N-1} \hat{x}_{m,m_i} \sum_n \frac{\partial L}{\partial Z_{n,n_i}} \hat{x}_{n,n_i} \right] \end{aligned}$$